

Four evidence layers across three research questions

EVIDENCE LAYER

QUESTION ADDRESSED

1 DELB: integer-MSE floor

Population bound and attainability;
paired estimated floors in application



RQ1 + RQ3

2 CMI + randomization

Added-information estimate; comparison
with design-matched null references



RQ2 + RQ3

3 Known-truth experiments

Known CMI, replicate estimates, bias,
power and detection limits



RQ2

4 Held-out MSE + Gap

Matched model errors and DELB;
realized use of prediction opportunity



RQ3

Layer 3 assesses behavior under tested designs; it is not a city-specific test.